Updates Since Preliminary Result: BDT, NPB Removal, Double Interactions, and Shower Shape Comparisons

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April 9, 2026

Abstract

This report summarizes the major analysis updates implemented since the preliminary isolated photon cross-section result in p+p collisions at $\sqrt{s} = 200$ GeV with sPHENIX Run 24 data. Four work streams are covered: (1) the photon identification BDT, expanded from a minimal feature set to 11 model variants with an E_{T}-dependent parametric threshold; (2) non-physics background (NPB) rejection, upgraded from a hard timing cut to a dedicated 25-feature XGBoost score; (3) double-interaction efficiency studies, demonstrating that the apparent reconstruction efficiency drop in pileup MC is a truth–reco matching artifact; and (4) shower shape data–MC comparisons using pileup-blended MC templates, which improve χ^2/ndf by factors of up to 22×. Each section documents the methodology, key results, and implications for the cross-section measurement.

Contents

1 Introduction **3**

2 BDT Photon Identification **3**
2.1 Feature Expansion . . . 3
2.2 Model Variants . . . 3
2.3 Training Procedure . . . 4
2.4 E_{T}-Dependent Parametric Threshold . . . 5
2.5 BDT Systematic Variations . . . 5

3 Non-Physics Background Rejection **5**
3.1 From Hard Cut to BDT Score . . . 5
3.2 NPB Score Training . . . 5
3.3 NPB Score Validation . . . 6
3.4 Cut Optimization . . . 6
3.5 NPB Systematic Variations . . . 6

4 Double-Interaction Efficiency Study **7**
4.1 Motivation . . . 7
4.2 Pileup Fraction Calculation . . . 7
4.3 Truth Photon Content . . . 7
4.4 Vertex Shift Mechanism . . . 8
4.5 Decomposition of the Efficiency Loss . . . 8
4.6 Per-Component Efficiency Breakdown . . . 9
4.7 `TEfficiency` Weighted-Event Fix . . . 9
4.8 Implications . . . 9

5 Shower Shape Data–MC Comparisons with Pileup Blending **9**
5.1 Two-Pass Blending Pipeline . . . 9
5.2 Comparison Methodology . . . 10
5.3 Results: 0 mrad Period . . . 10
5.3.1 w_η^{CoG} (Eta Cluster Width) . . . 10
5.3.2 BDT Score . . . 11
5.3.3 f_1 (Inner Ring Energy Fraction) . . . 12
5.3.4 Single vs. Double Interaction Overlay . . . 13
5.4 Results: 1.5 mrad Period . . . 13
5.5 Discussion . . . 14

6 Summary **14**

1 Introduction

The PPG12 isolated photon cross-section measurement uses sPHENIX Run 24 p+p data at $\sqrt{s} = 200$ GeV with an integrated luminosity of 16.6 pb^{-1} (1.5 mrad crossing-angle period, runs 51274–54000). The measurement employs the ABCD double-sideband method for background subtraction, with photon identification based on electromagnetic shower shape variables evaluated by a Boosted Decision Tree (BDT), and topological cluster isolation with $R = 0.4$. Twelve E_{T} bins span 8–35 GeV within $|\eta| < 0.7$.

Since the preliminary result, four areas have undergone substantial development:

1. **BDT photon identification** (§2): Feature expansion, model variant studies, parametric threshold, and systematic framework.
2. **NPB rejection** (§3): New dedicated XGBoost model replacing the hard timing preselection.
3. **Double-interaction efficiency** (§4): Decomposition of the pileup efficiency drop into matching artifact vs. genuine physics effects.
4. **Shower shape comparisons** (§5): Pileup-blended MC templates and quantitative data–MC agreement assessment.

2 BDT Photon Identification

2.1 Feature Expansion

The preliminary BDT used a minimal feature set of 6 variables: cluster E_{T}, η, ϕ, vertex z, $E_{1\times1}/E_{3\times3}$, and $E_{3\times2}/E_{3\times5}$. The updated analysis expands the training feature space to 25 variables, adding 13 tower energy ratios and width moments:

- **Tower energy fractions**: $e_{t1} = (E_1{+}E_2{+}E_3{+}E_4)/E_{\mathrm{tot}}$, e_{t2}, e_{t3}, e_{t4} (quadrant asymmetries)
- **Cluster width moments**: w_η^{CoG} (eta width), w_ϕ^{CoG} (phi width), $w_{3\times2}$, $w_{5\times2}$, $w_{7\times2}$
- **Energy ratios**: $E_{1\times1}/E_{2\times2}$, $E_{1\times1}/E_{1\times3}$, $E_{1\times1}/E_{1\times5}$, $E_{1\times1}/E_{1\times7}$, $E_{1\times1}/E_{3\times1}$, $E_{1\times1}/E_{5\times1}$, $E_{1\times1}/E_{7\times1}$, $E_{2\times2}/E_{3\times3}$, $E_{2\times2}/E_{3\times5}$, $E_{2\times2}/E_{3\times7}$, $E_{2\times2}/E_{5\times3}$

Training uses PYTHIA-8 photon MC (photon5/10/20, truth pid $\in \{1, 2\}$) as signal and jet MC (jet10/15/20/30/50, pid $\notin \{1, 2\}$) as background, with $6 < E_{\mathrm{T}} < 40$ GeV and $|\eta| < 0.7$.

2.2 Model Variants

To study feature importance and optimize the trade-off between discrimination power and robustness, 11 model variants are trained via systematic feature ablation:

Table 1: BDT model variants defined by feature subsets. All variants include vertex z, η, $E_{1\times1}/E_{3\times3}$, and e_{t1}–e_{t4} as baseline features. The "E" suffix denotes variants that include cluster E_{T} as an explicit feature.

Variant	Additional features	Count	Purpose
`base`	(baseline only)	7	Minimal reference
`base_vr`	(vertex-reweighted)	7	Vertex robustness check
`base_v0`	Drops e_{t1}	6	Remove isolation-correlated feature
`base_v1`	$+\,w_{\eta}^{\mathrm{CoG}}$	8	Add width
`base_v2`	$+\,w_{\eta}^{\mathrm{CoG}}$, w_{ϕ}^{CoG}	9	Add 2D shape
`base_v3`	$+\,w_{\eta}^{\mathrm{CoG}}$, w_{ϕ}^{CoG}, $E_{3\times2}/E_{3\times5}$	10	Full shape
`base_E`	$+\,E_{\mathrm{T}}$	8	Energy-aware base
`base_v0E`	$+\,E_{\mathrm{T}}$, drops e_{t1}	7	Energy + iso-insensitive
`base_v1E`	$+\,E_{\mathrm{T}}$, w_{η}^{CoG}	9	**Nominal**
`base_v2E`	$+\,E_{\mathrm{T}}$, w_{η}^{CoG}, w_{ϕ}^{CoG}	10	Energy + 2D shape
`base_v3E`	$+\,E_{\mathrm{T}}$, w_{η}^{CoG}, w_{ϕ}^{CoG}, $E_{3\times2}/E_{3\times5}$	11	Full feature set

The nominal analysis uses `base_v1E` (9 features) in both E_{T} bins $[8, 15]$ and $[15, 35]$ GeV, with `base_E` as the default fallback.

2.3 Training Procedure

The XGBoost training applies three-stage reweighting to minimize kinematic biases:

1. **Class reweighting**: Balances signal and background counts.
2. **E_{T} spectrum flattening**: Inverse-PDF spline reweighting to remove the steeply falling E_{T} spectrum, preventing the BDT from learning E_{T}-dependent shortcuts.
3. **η spectrum flattening**: Same procedure for the η distribution.

Background subsampling is applied below $E_{\mathrm{T}} = 15$ GeV to address the heavy-tail problem in the low-E_{T} region, using inverse-PDF reweighting across 20 E_{T} bins.

Table 2: XGBoost hyperparameters for the photon identification BDT.

Parameter	Value
Number of boosted trees	750
Maximum depth	5
Learning rate	0.1
Subsample fraction	0.5
Column sample per tree	0.6
L1 regularization (α)	5.0
L2 regularization (λ)	0.3
Grow policy	Loss-guided
Tree method	Histogram-based
Train / validation / test split	70% / 10% / 20%
Random seed	42

A split-sample approach divides MC into two independent halves to avoid self-training bias: each half is scored by the model trained on the other half. Models are exported via `TMVA::Experimental::SaveXGBoost` and applied in the analysis via `TMVA::Experimental::RBDT::Compute()`.

2.4 E_{T}-Dependent Parametric Threshold

The preliminary analysis used a flat BDT threshold. The updated analysis implements an E_{T}-dependent parametric cut:

$$\mathrm{BDT}_{\mathrm{min}}^{\mathrm{tight}} = 0.80 - 0.015 \times E_{\mathrm{T}}\,[\text{ GeV}], \tag{1}$$

where the intercept and slope are configurable in the analysis YAML. This accommodates the improving signal-to-background ratio at higher E_{T}, allowing a looser BDT requirement where the background contamination is naturally lower.

Clusters with $E_{\mathrm{T}} < 7$ GeV receive a default BDT score of -1 and are excluded from the analysis.

2.5 BDT Systematic Variations

The systematic uncertainty framework for the BDT is implemented through automated config generation (`make_bdt_variations.py`), producing 30+ analysis configurations. Key BDT-related variations:

Table 3: BDT-related systematic variations. Each variation generates a separate analysis config with a unique `var_type` suffix.

Variation	Parameter change	Syst. type	Role
`tightbdt50`	Intercept $\rightarrow$ 0.70	Efficiency	Down
`tightbdt70`	Intercept $\rightarrow$ 0.90	Efficiency	Up
`ntbdtmin05`	Non-tight lower $\rightarrow$ 0.05	Purity	One-sided
`ntbdtmin10`	Non-tight lower $\rightarrow$ 0.10	Purity	One-sided
`etbin_v3E_v3E`	Both bins use `base_v3E`	Efficiency	Max-envelope
`etbin_E_E`	Both bins use `base_E`	Efficiency	Max-envelope
`etbin_v1E_v3E`	Mixed models	Efficiency	Max-envelope

The tight BDT threshold variation yields 5–10% relative systematic uncertainty on the cross-section. The non-tight lower boundary variation contributes up to 15% relative uncertainty, primarily affecting the purity estimate through the ABCD sideband definition.

3 Non-Physics Background Rejection

3.1 From Hard Cut to BDT Score

The preliminary analysis rejected non-physics background (NPB) clusters using a hard timing preselection: clusters with $t_{\mathrm{cluster}} - t_{\mathrm{MBD}} < -5$ ns were tagged as NPB and removed. This approach has two limitations: (1) the timing threshold is somewhat arbitrary, and (2) it provides no discrimination for NPB clusters that happen to arrive near $t = 0$.

The updated analysis replaces this with a dedicated 25-feature XGBoost model that scores each cluster on a continuous $[0, 1]$ scale, where high scores indicate physics clusters and low scores indicate NPB.

3.2 NPB Score Training

The NPB BDT is trained on:

- **Signal (physics)**: Clusters from PYTHIA jet MC (jet10, jet15, jet20, jet30, jet50) with valid particle ID, $6 < E_{\mathrm{T}} < 40$ GeV, $|\eta| < 0.7$.

- **Background (NPB)**: Clusters from collision data with $t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns, which are overwhelmingly beam-induced backgrounds not synchronized with the collision time.

The model uses 25 input features: cluster E_{T}, η, vertex z, ten tower energy ratios ($E_{1\times1}/E_{3\times3}$, $E_{3\times2}/E_{3\times5}$, $E_{1\times1}/E_{2\times2}$, $E_{1\times1}/E_{1\times3}$, etc.), cluster width moments (w_{η}^{CoG}, w_{ϕ}^{CoG}, $w_{3\times2}$, $w_{5\times2}$, $w_{7\times2}$), and tower energy fractions (e_{t1}–e_{t4}).

Table 4: XGBoost hyperparameters for the NPB score model. Note the differences from the photon ID BDT (Table 2): deeper trees and less aggressive regularization, reflecting the cleaner signal–background separation for NPB rejection.

Parameter	Value
Number of boosted trees	750
Maximum depth	6
Learning rate	0.1
Subsample fraction	0.8
Column sample per tree	0.8
L1 regularization (α)	1.0
L2 regularization (λ)	1.0

The trained model is exported to ROOT TMVA format and applied in `apply_BDT.C`, producing a new branch `cluster_npb_score_CLUSTERINFO_CEMC` for each cluster with $6 < E_{\text{T}} < 40$ GeV and $|\eta| < 0.7$.

3.3 NPB Score Validation

The NPB score is validated independently using the cluster–MBD timing distribution $t_{\text{cluster}} - t_{\text{MBD}}$:

- At NPB score > 0.5: the timing distribution is sharply peaked near $t = 0$ ns, confirming physics origin.
- At NPB score < 0.2: the distribution is broad, extending to large negative times ($t < -7$ ns), confirming NPB origin.

A background template is constructed from the out-of-time tail ($t < -7$ ns) and normalized to estimate the NPB fraction as a function of the score threshold.

3.4 Cut Optimization

The nominal threshold is set at NPB score > 0.5, achieving:

- **NPB purity**: $\sim$99% of surviving clusters are physics (NPB contamination $\sim$1–2%).
- **Signal retention**: $\sim$98% of MC physics clusters pass the threshold.

This represents a significant improvement over the hard timing cut, which had less flexibility in balancing purity and efficiency.

3.5 NPB Systematic Variations

The NPB score threshold is varied as a systematic:

- `npb03`: Threshold $\rightarrow 0.3$ (looser, higher NPB contamination)
- `npb07`: Threshold $\rightarrow 0.7$ (tighter, lower signal retention)

The cross-section deviations from the nominal are taken as bin-by-bin systematic uncertainties.

4 Double-Interaction Efficiency Study

4.1 Motivation

When double-interaction MC samples (produced with full GEANT4 simulation of two overlapping p+p collisions) are passed through the standard efficiency pipeline, the reconstruction efficiency drops dramatically: from ∼94% (single interaction) to ∼31% (double interaction). This section documents a dedicated investigation demonstrating that this drop is overwhelmingly a truth–reco matching artifact caused by vertex displacement, not a genuine loss of signal photon clusters.

4.2 Pileup Fraction Calculation

The mean number of collisions per bunch crossing is computed from the Poisson-corrected MBD coincidence rate:

$$\mu = \frac{\sum_{\text{runs}} N_{\text{MBD,corr}}}{r_B \cdot \sum_{\text{runs}} \Delta t}, \tag{2}$$

where $r_B = 111 \times 78{,}200 = 8.68 \times 10^6$ Hz is the RHIC bunch crossing rate. The event-level interaction fractions are:

$$f_1 = \frac{\mu\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_2 = \frac{(\mu^2/2)\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_{\geq 3} = 1 - f_1 - f_2. \tag{3}$$

Fractions are computed per run and luminosity-weighted (not simple-averaged) to avoid Jensen's inequality bias. Cluster-weighted fractions account for the higher cluster multiplicity in multi-interaction events:

$$w_{\text{single}} = \frac{f_1}{\sum_k k\, f_k}, \qquad w_{\text{double}} = \frac{2\, f_2 + 3\, f_{\geq 3}}{\sum_k k\, f_k}. \tag{4}$$

Table 5: Luminosity-weighted pileup fractions and cluster-level mixing weights.

Crossing angle	Run range	μ_{corr}	f_1	f_2	w_{single}	w_{double}
0 mrad	47289–51274	0.24	0.879	0.111	0.776	0.224
1.5 mrad	51274–54000	0.15	0.930	0.066	0.867	0.133

4.3 Truth Photon Content

A prerequisite for interpreting the efficiency drop is verifying that the double-interaction MC does not inject additional truth-level prompt photons from the second collision.

Table 6: Truth-level prompt photon content per event (photon10 samples, 100k events each). The denominator requires $|\eta^\gamma| < 0.7$ and $E_{\text{T}}^{\text{iso,truth}} < 4$ GeV.

Quantity	Single MC	Double MC
Mean truth photons per event	0.493	0.495
Events with ≥ 2 denom. photons	0.22%	0.21%
Mean truth $E_{\text{T}}^{\text{iso,truth}}$ [GeV]	0.17	0.26

The truth photon content is statistically identical. The slightly higher mean truth isolation energy in double MC (0.26 vs. 0.17 GeV) reflects additional hadrons from the second collision falling within the isolation cone, but this is well below the 4 GeV fiducial threshold.

4.4 Vertex Shift Mechanism

In double-interaction events, the reconstructed vertex is the average of the two collision points:

$$z_{\text{vtx}}^{\text{reco}} = \frac{z_{\text{vtx},1} + z_{\text{vtx},2}}{2}. \tag{5}$$

Since both vertices are drawn from a broad distribution (RMS ∼30 cm), the displacement $\Delta z = z_{\text{vtx}}^{\text{reco}} - z_{\text{vtx}}^{\text{truth}}$ can be large. The cluster pseudorapidity shift at EMCal radius $R_{\text{EMCal}} = 93.5$ cm is:

$$\Delta\eta \approx -\frac{\Delta z}{R_{\text{EMCal}} \cosh \eta}, \tag{6}$$

which at $\eta = 0$ gives $|\Delta\eta| \approx |\Delta z|/93.5$ cm. The $\Delta R < 0.1$ matching threshold therefore fails when $|\Delta z| \gtrsim 9.35$ cm at midrapidity.

Table 7: Vertex shift distributions. The critical threshold $|\Delta z| = 9.35$ cm corresponds to $|\Delta\eta| = 0.1$ at $\eta = 0$.

Quantity	Single MC	Double MC
RMS of Δz [cm]	7.6	34.5
Fraction with $|\Delta z| > 9.35$ cm	—	83%

The sharp transition in matching failure rate as a function of $|\Delta z|$ confirms the deterministic geometric origin:

Table 8: Matching failure rate vs. vertex displacement in double MC.

$|\Delta z|$ range [cm]	Matching failure rate
< 10	9%
10–15	77%
15–20	∼100%
> 20	100%

4.5 Decomposition of the Efficiency Loss

For truth photons that fail the $\Delta R < 0.1$ match in double MC, the failure mode is classified by searching for the `trkID`-matched cluster regardless of ΔR:

- **ΔR failure (88%)**: A cluster with the correct `trkID` exists but $\Delta R > 0.1$ due to vertex shift. The signal photon cluster is present and intact — it merely appears at the wrong η when reconstructed from the displaced vertex.
- **Missing cluster (12%)**: No cluster with the matching `trkID` is found, corresponding to genuine cluster loss from merging in the higher-multiplicity environment.
- **Extra truth photons (∼0%)**: The second collision does not produce additional prompt photons.

The 88% ΔR-failure fraction, combined with the 63-point overall efficiency drop (94% → 31%), means ∼55 percentage points are attributable to the vertex-shift artifact and only ∼8 points to genuine cluster loss.

4.6 Per-Component Efficiency Breakdown

Table 9: Per-component efficiency comparison, averaged over $E_T^\gamma = 10$–35 GeV. Ranges indicate variation across E_T^γ bins.

Efficiency component	Single MC	Double MC	Effect
Reco matching ($\Delta R < 0.1$)	93–94%	27–32%	Dominant (artifact)
Photon ID (BDT)	81–92%	81–92%	**Negligible**
Isolation (E_T^{iso})	61–74%	58–71%	∼4% degradation
All combined	52–57%	15–19%	Dominated by reco

The photon ID (BDT) efficiency is **identical** in single and double MC, demonstrating that the shower-shape BDT is robust against pileup. The cluster retains its electromagnetic shower profile even when the vertex is displaced. The isolation efficiency shows a modest ∼4% degradation from additional energy deposited by the second collision within the isolation cone.

4.7 `TEfficiency` Weighted-Event Fix

During this investigation, a bug was identified in the `TEfficiency` usage within `RecoEffCalculator_TTreeReade`
The original `SetWeight(w) + Fill(pass, pT)` pattern stores unweighted event counts, biasing the merged efficiency toward the sample with the most raw events. The corrected implementation uses `SetUseWeightedEvents()` with `FillWeighted(pass, weight, pT)`, which stores weighted sums so that `hadd` correctly combines samples by physics weight.

4.8 Implications

The central finding is that the apparent reconstruction efficiency drop in double-interaction MC is a matching artifact, not a physics inefficiency. For the nominal 1.5 mrad period with 13.3% cluster-weighted pileup:

- **Isolation degradation**: ∼4% in double MC × 13.3% pileup fraction = ∼0.5% overall effect.
- **BDT efficiency**: Unaffected — shower shapes are pileup-robust.
- **Genuine cluster loss**: ∼8% of double-MC truth photons × 13.3% = ∼1% overall.

The efficiency correction uses single-interaction MC only. Double-interaction MC is reserved for shower-shape template comparisons (§5).

5 Shower Shape Data–MC Comparisons with Pileup Blending

5.1 Two-Pass Blending Pipeline

The pileup mixing uses a two-pass procedure with `ShowerShapeCheck.C`:

Pass 1 (vertex scan): The z_{vtx} distribution is filled for each MC sample weighted by the cluster-level fraction (w_{single} or w_{double} from Table 5). The resulting histograms are merged via `hadd` to produce a blended MC vertex distribution.

Pass 2 (full analysis): All MC samples are reprocessed using the blended vertex distribution from Pass 1 for vertex reweighting, ensuring consistent z_{vtx} templates. Each sample's fills are scaled by its mixing weight multiplicatively on top of the cross-section weight. Single and double outputs are merged to produce the final blended MC.

The MC samples are photon10 (signal, 14–30 GeV truth p_{T}) and jet12 (background, 14–21 GeV), each with a corresponding `_double` sample.

5.2 Comparison Methodology

Shower shape distributions are compared at two selection levels:

- **cut0**: No selection. Data includes NPB contamination not modeled in MC, complicating interpretation.
- **cut1**: After NPB preselection (NPB score > 0.5). Removes non-physics clusters, providing a cleaner pileup assessment. **This is the preferred comparison level.**

Each comparison plot shows data (black), signal MC (red), and inclusive MC (blue), area-normalized, with χ^2/ndf and a Data/Inclusive MC ratio panel.

5.3 Results: 0 mrad Period

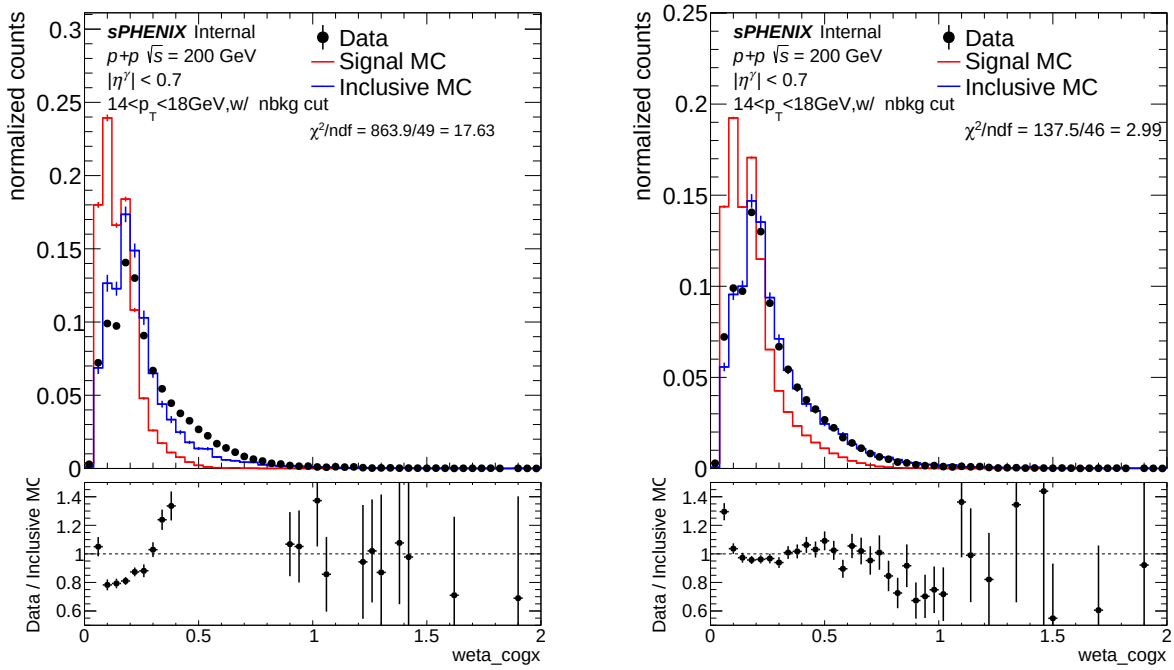
Table 10 compiles the χ^2/ndf values at cut1 for four representative shower shape variables, comparing nominal (single-only) and mixed (18.7% double-blended) MC.

Table 10: Data–Inclusive MC χ^2/ndf for nominal and mixed MC at cut1, 0 mrad period. Lower values indicate better agreement.

Variable	E_{T} bin [GeV]	Nominal	Mixed	Improvement
w_η^{CoG}	10–14 (pt0)	47.4	29.1	1.6×
w_η^{CoG}	14–18 (pt1)	17.6	3.0	5.9×
w_η^{CoG}	22–28 (pt3)	16.9	0.75	22.5×
BDT score	14–18 (pt1)	12.2	1.6	7.5×
f_1 (inner ring)	14–18 (pt1)	9.3	1.9	4.8×
$E_{1\times1}/E_{3\times3}$	14–18 (pt1)	4.5	4.1	1.1×

5.3.1 w_η^{CoG} (Eta Cluster Width)

w_η^{CoG} is the variable most sensitive to double interactions: vertex shifts displace clusters along the beam axis, broadening the η profile. The improvement grows with E_{T}: at 22–28 GeV, χ^2/ndf drops from 16.9 to 0.75 (22.5×), indicating that the nominal MC's failure to model pileup-induced broadening becomes increasingly significant relative to statistical uncertainties.



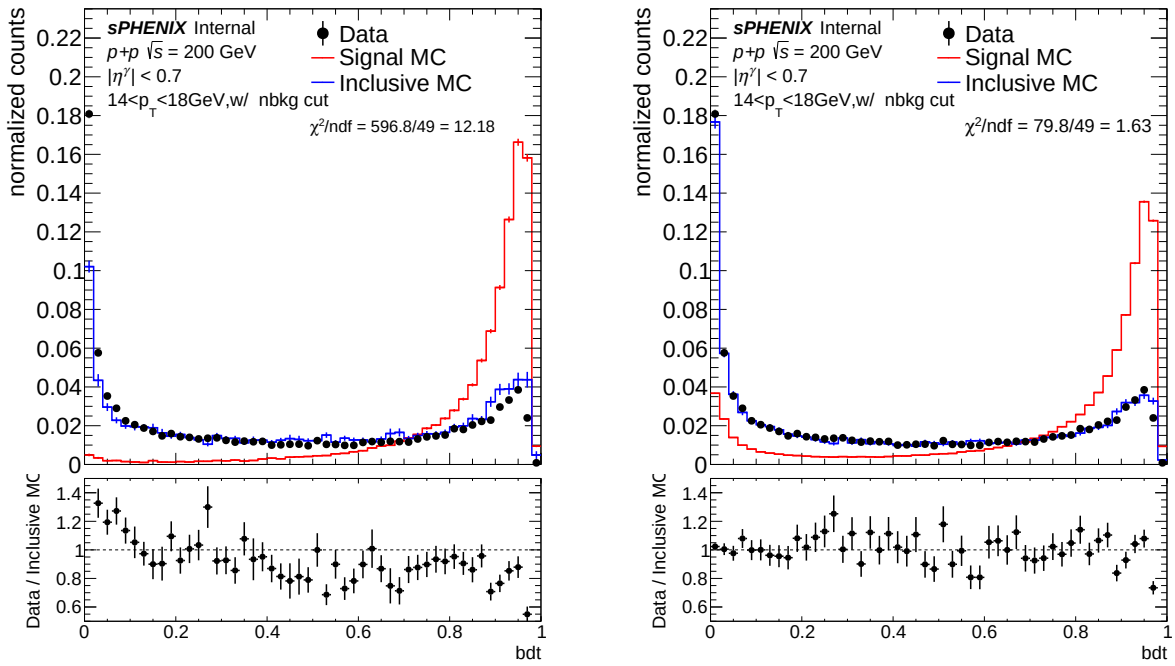
(a) Nominal MC

(b) Mixed MC

Figure 1: w_η^{CoG} at $14 < E_{\mathrm{T}} < 18$ GeV, 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 17.6$; mixed $\chi^2/\mathrm{ndf} = 3.0$ (5.9× improvement).

5.3.2 BDT Score

The BDT score amplifies underlying shower shape distortions through the XGBoost decision function, making it the most sensitive composite variable.



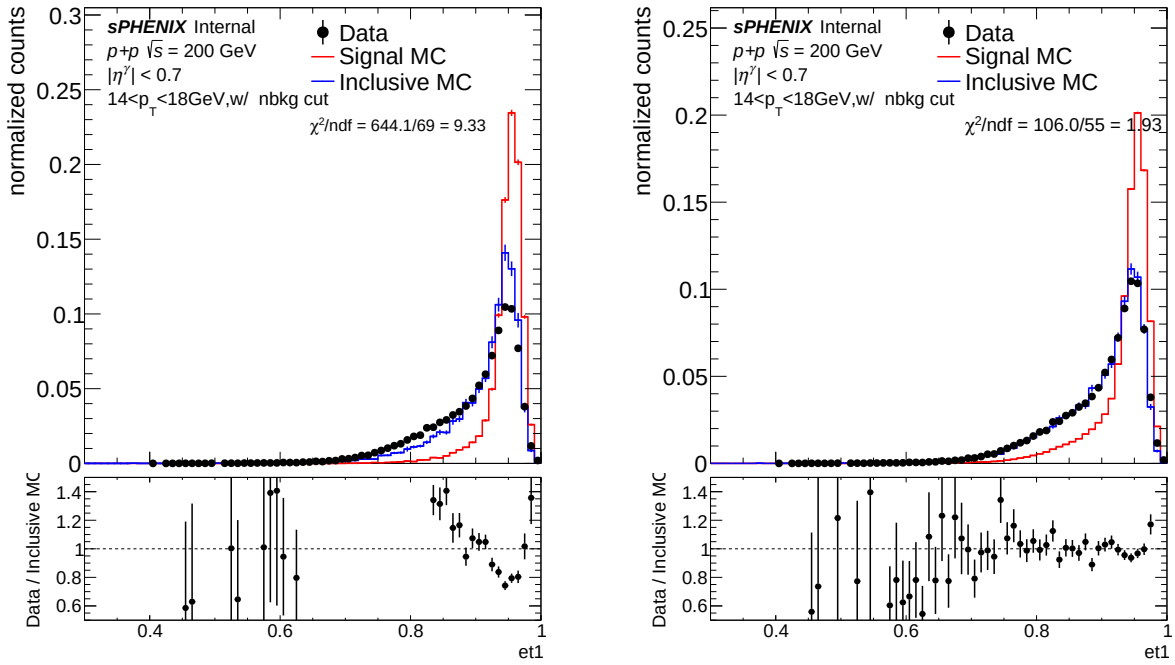
(a) Nominal MC

(b) Mixed MC

Figure 2: BDT score at $14 < E_{\mathrm{T}} < 18$ GeV, 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 12.2$; mixed $\chi^2/\mathrm{ndf} = 1.6$ (7.5× improvement).

5.3.3 f_1 (Inner Ring Energy Fraction)

Double interactions spread energy outward from the shower core, reducing f_1.



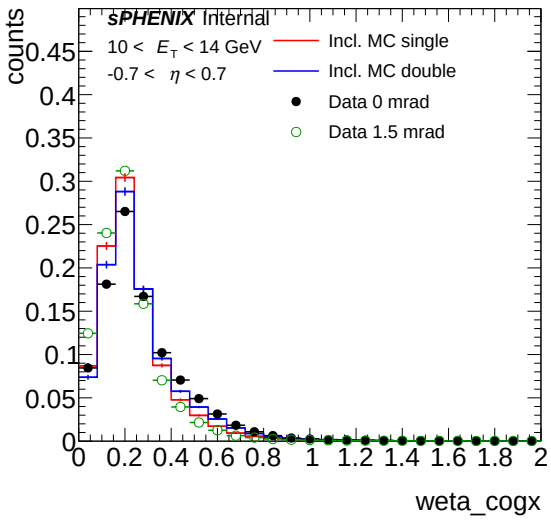
(a) Nominal MC

(b) Mixed MC

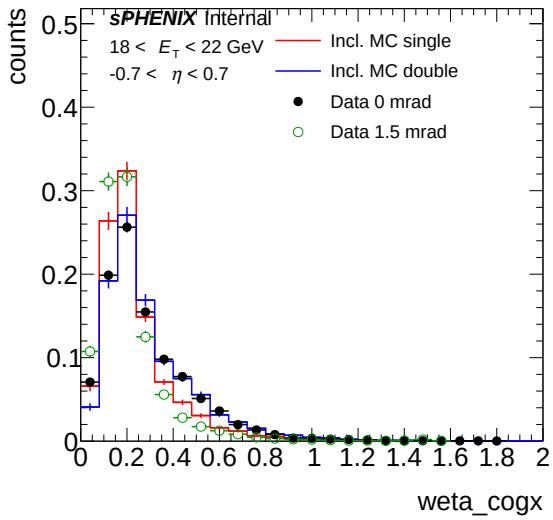
Figure 3: f_1 at $14 < E_{\mathrm{T}} < 18$ GeV, 0 mrad, cut1. Nominal $\chi^2/\mathrm{ndf} = 9.3$; mixed $\chi^2/\mathrm{ndf} = 1.9$ (4.8× improvement).

5.3.4 Single vs. Double Interaction Overlay

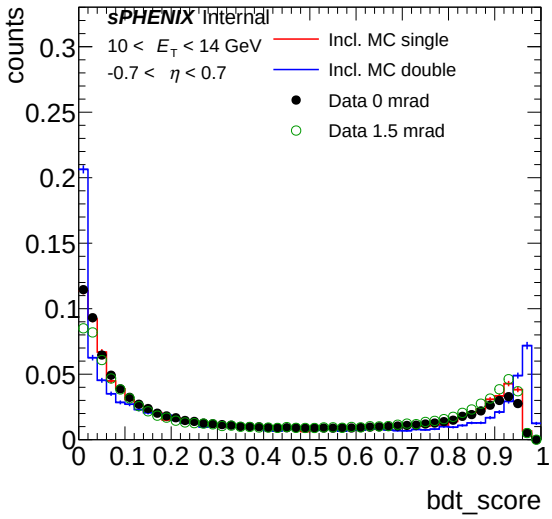
Figure 4 overlays single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods. Data at 0 mrad systematically tracks the blended expectation (closer to the double MC tail), while data at 1.5 mrad tracks closer to single MC, consistent with the different pileup fractions.



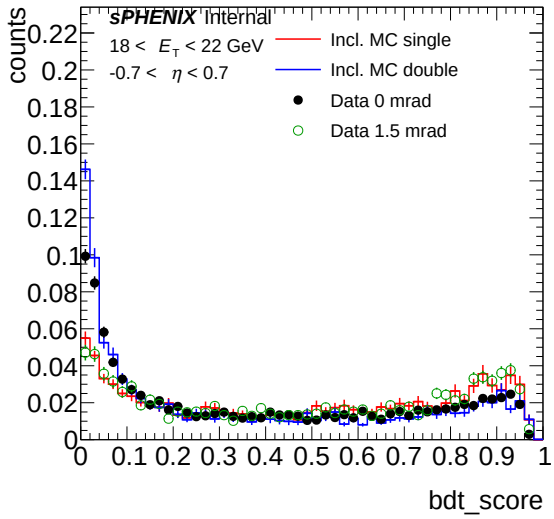
(a) w_η^{CoG}, 10–14 GeV



(b) w_η^{CoG}, 18–22 GeV



(c) BDT score, 10–14 GeV



(d) BDT score, 18–22 GeV

Figure 4: Shower shape distributions: single MC (red), double MC (blue), data 0 mrad (filled black), data 1.5 mrad (open green). Left: $10 < E_{\mathrm{T}} < 14$ GeV. Right: $18 < E_{\mathrm{T}} < 22$ GeV.

5.4 Results: 1.5 mrad Period

The 1.5 mrad period (nominal dataset) has a smaller double-interaction fraction (13.3% cluster-weighted), so the mixing effect is correspondingly smaller. The improvement pattern is consistent with 0 mrad but reduced in magnitude, confirming that the effect scales with the pileup fraction.

5.5 Discussion

The mixed MC dramatically improves data–MC agreement for pileup-sensitive variables (w_η^{CoG} and BDT score), with χ^2/ndf improvement factors of 1.6× to 22.5×. The improvement grows with E_{T}: at 22–28 GeV, the nominal MC discrepancy is most systematic relative to statistical uncertainties, and the mixing provides the largest relative correction.

The $E_{1\times1}/E_{3\times3}$ ratio shows only 1.1× improvement, indicating this variable is less sensitive to pileup-induced broadening or that competing effects partially cancel.

For the nominal cross-section measurement using the 1.5 mrad period, the pileup mixing is **not applied as a correction** because:

1. The pileup fraction is small (13.3% cluster-weighted).
2. The BDT-based photon identification is unaffected by pileup (Table 9).
3. The effect is within existing systematic uncertainties.

The study serves as a cross-check that unmodeled pileup does not significantly bias the measurement.

6 Summary

Table 11 provides a concise overview of the four work streams and their impact on the cross-section measurement.

Table 11: Summary of updates since the preliminary result.

Work stream	Key change	Result	Cross-section impact
BDT photon ID	6 → 25 features, 11 model variants, E_{T}-dependent threshold	Systematic framework with 30+ configs; 5–10% uncertainty from threshold	Improved discrimination and well-characterized systematics
NPB rejection	Hard timing cut → 25-feature XGBoost score	∼99% purity at ∼98% signal retention	Cleaner data with flexible threshold systematics
Double interaction efficiency	Decomposition of 63-pt efficiency drop	88% is matching artifact; BDT efficiency unaffected	Single-MC efficiency validated; $<$ 1% overall pileup effect
Shower shape comparisons	Pileup-blended MC templates	χ^2/ndf improvement up to 22×	Cross-check confirms pileup within existing systematics

The combined conclusion is that:

1. The photon identification BDT is robust against pileup, with identical efficiency in single and double MC.
2. The NPB rejection provides cleaner data samples with well-controlled systematics.

3. The apparent efficiency drop in double-interaction MC is a truth–reco matching artifact, not a physics inefficiency.
4. Pileup-blended MC templates substantially improve data–MC shower shape agreement, validating the physics-motivated mixing fractions.
5. For the nominal 1.5 mrad measurement, pileup effects are sub-percent and within existing systematic uncertainties.